

lab23(SSU)

November 26, 2025

Helpful Resource:

- [Python Reference](#)

Recommended Readings:

- [Variability](#)
- [Correlation](#)

```
[1]: !apt-get install texlive texlive-xetex texlive-latex-extra pandoc
!pip install py pandoc
!pip install datascience
```

Reading package lists... Done

Building dependency tree... Done

Reading state information... Done

pandoc is already the newest version (2.9.2.1-3ubuntu2).

pandoc set to manually installed.

The following additional packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsyntax2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration t1utils
teckit tex-common tex-gyre texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-recommended
texlive-pictures texlive-plain-generic tipa xfonts-encodings xfonts-utils

Suggested packages:

fonts-noto fonts-freefont-otf | fonts-freefont-ttf libavalon-framework-java
libcommons-logging-java-doc libexcalibur-logkit-java liblog4j1.2-java
poppler-utils ghostscript fonts-japanese-mincho | fonts-ipafont-mincho
fonts-japanese-gothic | fonts-ipafont-gothic fonts-arphic-ukai
fonts-arphic-uming fonts-nanum ri ruby-dev bundler debhelper gv
| postscript-viewer perl-tk xpdf | pdf-viewer xzdec
texlive-fonts-recommended-doc texlive-latex-base-doc python3-pygments
icc-profiles libfile-which-perl libspreadsheet-parseexcel-perl
texlive-latex-extra-doc texlive-latex-recommended-doc texlive-luatex
texlive-pstricks dot2tex prerex texlive-pictures-doc vprerex

default-jre-headless tipa-doc

The following NEW packages will be installed:

dvisvgm fonts-droid-fallback fonts-lato fonts-lmodern fonts-noto-mono
fonts-texgyre fonts-urw-base35 libapache-pom-java libcommons-logging-java
libcommons-parent-java libfontbox-java libgs9 libgs9-common libidn12
libijs-0.35 libjbig2dec0 libkpathsea6 libpdfbox-java libptexenc1 libruby3.0
libsynctex2 libteckit0 libtexlua53 libtexluajit2 libwoff1 libzip-0-13
lmodern poppler-data preview-latex-style rake ruby ruby-net-telnet
ruby-rubygems ruby-webrick ruby-xmlrpc ruby3.0 rubygems-integration tiutils
teckit tex-common tex-gyre texlive texlive-base texlive-binaries
texlive-fonts-recommended texlive-latex-base texlive-latex-extra
texlive-latex-recommended texlive-pictures texlive-plain-generic
texlive-xetex tipa xfonts-encodings xfonts-utils

0 upgraded, 54 newly installed, 0 to remove and 41 not upgraded.

Need to get 182 MB of archives.

After this operation, 571 MB of additional disk space will be used.

Get:1 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-droid-fallback all 1:6.0.1r16-1.1build1 [1,805 kB]

Get:2 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-lato all 2.0-2.1 [2,696 kB]

Get:3 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 poppler-data all 0.4.11-1 [2,171 kB]

Get:4 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-common all 6.17 [33.7 kB]

Get:5 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-urw-base35 all 20200910-1 [6,367 kB]

Get:6 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9-common all 9.55.0~dfsg1-0ubuntu5.13 [753 kB]

Get:7 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libidn12 amd64 1.38-4ubuntu1 [60.0 kB]

Get:8 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libijs-0.35 amd64 0.35-15build2 [16.5 kB]

Get:9 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libjbig2dec0 amd64 0.19-3build2 [64.7 kB]

Get:10 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libgs9 amd64 9.55.0~dfsg1-0ubuntu5.13 [5,032 kB]

Get:11 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libkpathsea6 amd64 2021.20210626.59705-1ubuntu0.2 [60.4 kB]

Get:12 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 libwoff1 amd64 1.0.2-1build4 [45.2 kB]

Get:13 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 dvisvgm amd64 2.13.1-1 [1,221 kB]

Get:14 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-lmodern all 2.004.5-6.1 [4,532 kB]

Get:15 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 fonts-noto-mono all 20201225-1build1 [397 kB]

Get:16 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 fonts-texgyre all 20180621-3.1 [10.2 MB]

Get:17 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libapache-pom-java
 all 18-1 [4,720 B]
 Get:18 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-parent-
 java all 43-1 [10.8 kB]
 Get:19 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libcommons-logging-
 java all 1.2-2 [60.3 kB]
 Get:20 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libptexenc1
 amd64 2021.20210626.59705-1ubuntu0.2 [39.1 kB]
 Get:21 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rubygems-integration
 all 1.18 [5,336 B]
 Get:22 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby3.0 amd64
 3.0.2-7ubuntu2.11 [50.1 kB]
 Get:23 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-rubygems
 all 3.3.5-2ubuntu1.2 [228 kB]
 Get:24 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby amd64 1:3.0~exp1
 [5,100 B]
 Get:25 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 rake all 13.0.6-2 [61.7
 kB]
 Get:26 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 ruby-net-telnet all
 0.1.1-2 [12.6 kB]
 Get:27 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-webrick
 all 1.7.0-3ubuntu0.2 [52.5 kB]
 Get:28 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 ruby-xmlrpc all
 0.3.2-1ubuntu0.1 [24.9 kB]
 Get:29 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libruby3.0
 amd64 3.0.2-7ubuntu2.11 [5,114 kB]
 Get:30 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libsinctex2
 amd64 2021.20210626.59705-1ubuntu0.2 [55.6 kB]
 Get:31 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libteckit0 amd64
 2.5.11+ds1-1 [421 kB]
 Get:32 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexlua53
 amd64 2021.20210626.59705-1ubuntu0.2 [120 kB]
 Get:33 <http://archive.ubuntu.com/ubuntu> jammy-updates/main amd64 libtexluajit2
 amd64 2021.20210626.59705-1ubuntu0.2 [267 kB]
 Get:34 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libzip-0-13 amd64
 0.13.72+dfsg.1-1.1 [27.0 kB]
 Get:35 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-encodings all
 1:1.0.5-0ubuntu2 [578 kB]
 Get:36 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 xfonts-utils amd64
 1:7.7+6build2 [94.6 kB]
 Get:37 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 lmodern all
 2.004.5-6.1 [9,471 kB]
 Get:38 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 preview-latex-style
 all 12.2-1ubuntu1 [185 kB]
 Get:39 <http://archive.ubuntu.com/ubuntu> jammy/main amd64 t1utils amd64
 1.41-4build2 [61.3 kB]
 Get:40 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 teckit amd64
 2.5.11+ds1-1 [699 kB]

Get:41 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tex-gyre all
 20180621-3.1 [6,209 kB]
 Get:42 <http://archive.ubuntu.com/ubuntu> jammy-updates/universe amd64 texlive-
 binaries amd64 2021.20210626.59705-1ubuntu0.2 [9,860 kB]
 Get:43 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-base all
 2021.20220204-1 [21.0 MB]
 Get:44 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-fonts-
 recommended all 2021.20220204-1 [4,972 kB]
 Get:45 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-base
 all 2021.20220204-1 [1,128 kB]
 Get:46 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-
 recommended all 2021.20220204-1 [14.4 MB]
 Get:47 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive all
 2021.20220204-1 [14.3 kB]
 Get:48 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libfontbox-java all
 1:1.8.16-2 [207 kB]
 Get:49 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 libpdfbox-java all
 1:1.8.16-2 [5,199 kB]
 Get:50 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-pictures
 all 2021.20220204-1 [8,720 kB]
 Get:51 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-latex-extra
 all 2021.20220204-1 [13.9 MB]
 Get:52 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-plain-
 generic all 2021.20220204-1 [27.5 MB]
 Get:53 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 tipa all 2:1.3-21
 [2,967 kB]
 Get:54 <http://archive.ubuntu.com/ubuntu> jammy/universe amd64 texlive-xetex all
 2021.20220204-1 [12.4 MB]
 Fetched 182 MB in 4s (47.8 MB/s)
 Extracting templates from packages: 100%
 Preconfiguring packages ...
 Selecting previously unselected package fonts-droid-fallback.
 (Reading database ... 121713 files and directories currently installed.)
 Preparing to unpack .../00-fonts-droid-fallback_1%3a6.0.1r16-1.1build1_all.deb
 ...
 Unpacking fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
 Selecting previously unselected package fonts-lato.
 Preparing to unpack .../01-fonts-lato_2.0-2.1_all.deb ...
 Unpacking fonts-lato (2.0-2.1) ...
 Selecting previously unselected package poppler-data.
 Preparing to unpack .../02-poppler-data_0.4.11-1_all.deb ...
 Unpacking poppler-data (0.4.11-1) ...
 Selecting previously unselected package tex-common.
 Preparing to unpack .../03-tex-common_6.17_all.deb ...
 Unpacking tex-common (6.17) ...
 Selecting previously unselected package fonts-urw-base35.
 Preparing to unpack .../04-fonts-urw-base35_20200910-1_all.deb ...
 Unpacking fonts-urw-base35 (20200910-1) ...

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Selecting previously unselected package libgs9-common.
Preparing to unpack .../05-libgs9-common_9.55.0~dfsg1-0ubuntu5.13_all.deb ...
Unpacking libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libidn12:amd64.
Preparing to unpack .../06-libidn12_1.38-4ubuntu1_amd64.deb ...
Unpacking libidn12:amd64 (1.38-4ubuntu1) ...
Selecting previously unselected package libijs-0.35:amd64.
Preparing to unpack .../07-libijs-0.35_0.35-15build2_amd64.deb ...
Unpacking libijs-0.35:amd64 (0.35-15build2) ...
Selecting previously unselected package libjbig2dec0:amd64.
Preparing to unpack .../08-libjbig2dec0_0.19-3build2_amd64.deb ...
Unpacking libjbig2dec0:amd64 (0.19-3build2) ...
Selecting previously unselected package libgs9:amd64.
Preparing to unpack .../09-libgs9_9.55.0~dfsg1-0ubuntu5.13_amd64.deb ...
Unpacking libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Selecting previously unselected package libkpathsea6:amd64.
Preparing to unpack .../10-libkpathsea6_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libwoff1:amd64.
Preparing to unpack .../11-libwoff1_1.0.2-1build4_amd64.deb ...
Unpacking libwoff1:amd64 (1.0.2-1build4) ...
Selecting previously unselected package dvisvgm.
Preparing to unpack .../12-dvisvgm_2.13.1-1_amd64.deb ...
Unpacking dvisvgm (2.13.1-1) ...
Selecting previously unselected package fonts-lmodern.
Preparing to unpack .../13-fonts-lmodern_2.004.5-6.1_all.deb ...
Unpacking fonts-lmodern (2.004.5-6.1) ...
Selecting previously unselected package fonts-noto-mono.
Preparing to unpack .../14-fonts-noto-mono_20201225-1build1_all.deb ...
Unpacking fonts-noto-mono (20201225-1build1) ...
Selecting previously unselected package fonts-texgyre.
Preparing to unpack .../15-fonts-texgyre_20180621-3.1_all.deb ...
Unpacking fonts-texgyre (20180621-3.1) ...
Selecting previously unselected package libapache-pom-java.
Preparing to unpack .../16-libapache-pom-java_18-1_all.deb ...
Unpacking libapache-pom-java (18-1) ...
Selecting previously unselected package libcommons-parent-java.
Preparing to unpack .../17-libcommons-parent-java_43-1_all.deb ...
Unpacking libcommons-parent-java (43-1) ...
Selecting previously unselected package libcommons-logging-java.
Preparing to unpack .../18-libcommons-logging-java_1.2-2_all.deb ...
Unpacking libcommons-logging-java (1.2-2) ...
Selecting previously unselected package libptexenc1:amd64.
Preparing to unpack .../19-libptexenc1_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package rubygems-integration.

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Preparing to unpack .../20-rubygems-integration_1.18_all.deb ...
Unpacking rubygems-integration (1.18) ...
Selecting previously unselected package ruby3.0.
Preparing to unpack .../21-ruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking ruby3.0 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package ruby-rubygems.
Preparing to unpack .../22-ruby-rubygems_3.3.5-2ubuntu1.2_all.deb ...
Unpacking ruby-rubygems (3.3.5-2ubuntu1.2) ...
Selecting previously unselected package ruby.
Preparing to unpack .../23-ruby_1%3a3.0~exp1_amd64.deb ...
Unpacking ruby (1:3.0~exp1) ...
Selecting previously unselected package rake.
Preparing to unpack .../24-rake_13.0.6-2_all.deb ...
Unpacking rake (13.0.6-2) ...
Selecting previously unselected package ruby-net-telnet.
Preparing to unpack .../25-ruby-net-telnet_0.1.1-2_all.deb ...
Unpacking ruby-net-telnet (0.1.1-2) ...
Selecting previously unselected package ruby-webrick.
Preparing to unpack .../26-ruby-webrick_1.7.0-3ubuntu0.2_all.deb ...
Unpacking ruby-webrick (1.7.0-3ubuntu0.2) ...
Selecting previously unselected package ruby-xmlrpc.
Preparing to unpack .../27-ruby-xmlrpc_0.3.2-1ubuntu0.1_all.deb ...
Unpacking ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Selecting previously unselected package libruby3.0:amd64.
Preparing to unpack .../28-libruby3.0_3.0.2-7ubuntu2.11_amd64.deb ...
Unpacking libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Selecting previously unselected package libsyntax2:amd64.
Preparing to unpack .../29-libsyntax2_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libsyntax2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libteckit0:amd64.
Preparing to unpack .../30-libteckit0_2.5.11+ds1-1_amd64.deb ...
Unpacking libteckit0:amd64 (2.5.11+ds1-1) ...
Selecting previously unselected package libtexlua53:amd64.
Preparing to unpack .../31-libtexlua53_2021.20210626.59705-1ubuntu0.2_amd64.deb
...
Unpacking libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libtexluaajit2:amd64.
Preparing to unpack
.../32-libtexluaajit2_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking libtexluaajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package libzip-0-13:amd64.
Preparing to unpack .../33-libzip-0-13_0.13.72+dfsg.1-1.1_amd64.deb ...
Unpacking libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Selecting previously unselected package xfonts-encodings.
Preparing to unpack .../34-xfonts-encodings_1%3a1.0.5-0ubuntu2_all.deb ...
Unpacking xfonts-encodings (1:1.0.5-0ubuntu2) ...
Selecting previously unselected package xfonts-utils.

```

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Preparing to unpack .../35-xfonts-utils_1%3a7.7+6build2_amd64.deb ...
Unpacking xfonts-utils (1:7.7+6build2) ...
Selecting previously unselected package lmodern.
Preparing to unpack .../36-lmodern_2.004.5-6.1_all.deb ...
Unpacking lmodern (2.004.5-6.1) ...
Selecting previously unselected package preview-latex-style.
Preparing to unpack .../37-preview-latex-style_12.2-1ubuntu1_all.deb ...
Unpacking preview-latex-style (12.2-1ubuntu1) ...
Selecting previously unselected package tlutils.
Preparing to unpack .../38-tlutils_1.41-4build2_amd64.deb ...
Unpacking tlutils (1.41-4build2) ...
Selecting previously unselected package teckit.
Preparing to unpack .../39-teckit_2.5.11+ds1-1_amd64.deb ...
Unpacking teckit (2.5.11+ds1-1) ...
Selecting previously unselected package tex-gyre.
Preparing to unpack .../40-tex-gyre_20180621-3.1_all.deb ...
Unpacking tex-gyre (20180621-3.1) ...
Selecting previously unselected package texlive-binaries.
Preparing to unpack .../41-texlive-
binaries_2021.20210626.59705-1ubuntu0.2_amd64.deb ...
Unpacking texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
Selecting previously unselected package texlive-base.
Preparing to unpack .../42-texlive-base_2021.20220204-1_all.deb ...
Unpacking texlive-base (2021.20220204-1) ...
Selecting previously unselected package texlive-fonts-recommended.
Preparing to unpack .../43-texlive-fonts-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-fonts-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-base.
Preparing to unpack .../44-texlive-latex-base_2021.20220204-1_all.deb ...
Unpacking texlive-latex-base (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-recommended.
Preparing to unpack .../45-texlive-latex-recommended_2021.20220204-1_all.deb ...
Unpacking texlive-latex-recommended (2021.20220204-1) ...
Selecting previously unselected package texlive.
Preparing to unpack .../46-texlive_2021.20220204-1_all.deb ...
Unpacking texlive (2021.20220204-1) ...
Selecting previously unselected package libfontbox-java.
Preparing to unpack .../47-libfontbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libfontbox-java (1:1.8.16-2) ...
Selecting previously unselected package libpdfbox-java.
Preparing to unpack .../48-libpdfbox-java_1%3a1.8.16-2_all.deb ...
Unpacking libpdfbox-java (1:1.8.16-2) ...
Selecting previously unselected package texlive-pictures.
Preparing to unpack .../49-texlive-pictures_2021.20220204-1_all.deb ...
Unpacking texlive-pictures (2021.20220204-1) ...
Selecting previously unselected package texlive-latex-extra.
Preparing to unpack .../50-texlive-latex-extra_2021.20220204-1_all.deb ...
Unpacking texlive-latex-extra (2021.20220204-1) ...

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Selecting previously unselected package texlive-plain-generic.
Preparing to unpack .../51-texlive-plain-generic_2021.20220204-1_all.deb ...
Unpacking texlive-plain-generic (2021.20220204-1) ...
Selecting previously unselected package tipa.
Preparing to unpack .../52-tipa_2%3a1.3-21_all.deb ...
Unpacking tipa (2:1.3-21) ...
Selecting previously unselected package texlive-xetex.
Preparing to unpack .../53-texlive-xetex_2021.20220204-1_all.deb ...
Unpacking texlive-xetex (2021.20220204-1) ...
Setting up fonts-lato (2.0-2.1) ...
Setting up fonts-noto-mono (20201225-1build1) ...
Setting up libwoff1:amd64 (1.0.2-1build4) ...
Setting up libtexlua53:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libijs-0.35:amd64 (0.35-15build2) ...
Setting up libtexluajit2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libfontbox-java (1:1.8.16-2) ...
Setting up rubygems-integration (1.18) ...
Setting up libzip-0-13:amd64 (0.13.72+dfsg.1-1.1) ...
Setting up fonts-urw-base35 (20200910-1) ...
Setting up poppler-data (0.4.11-1) ...
Setting up tex-common (6.17) ...
update-language: texlive-base not installed and configured, doing nothing!
Setting up libbig2dec0:amd64 (0.19-3build2) ...
Setting up libteckit0:amd64 (2.5.11+ds1-1) ...
Setting up libapache-pom-java (18-1) ...
Setting up ruby-net-telnet (0.1.1-2) ...
Setting up xfonts-encodings (1:1.0.5-0ubuntu2) ...
Setting up t1utils (1.41-4build2) ...
Setting up libidn12:amd64 (1.38-4ubuntu1) ...
Setting up fonts-texgyre (20180621-3.1) ...
Setting up libkpathsea6:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up ruby-webrick (1.7.0-3ubuntu0.2) ...
Setting up fonts-lmodern (2.004.5-6.1) ...
Setting up fonts-droid-fallback (1:6.0.1r16-1.1build1) ...
Setting up ruby-xmlrpc (0.3.2-1ubuntu0.1) ...
Setting up libsynchronet2:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up libgs9-common (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up teckit (2.5.11+ds1-1) ...
Setting up libpdfbox-java (1:1.8.16-2) ...
Setting up libgs9:amd64 (9.55.0~dfsg1-0ubuntu5.13) ...
Setting up preview-latex-style (12.2-1ubuntu1) ...
Setting up libcommons-parent-java (43-1) ...
Setting up dvisvgm (2.13.1-1) ...
Setting up libcommons-logging-java (1.2-2) ...
Setting up xfonts-utils (1:7.7+6build2) ...
Setting up libptexenc1:amd64 (2021.20210626.59705-1ubuntu0.2) ...
Setting up texlive-binaries (2021.20210626.59705-1ubuntu0.2) ...
update-alternatives: using /usr/bin/xdvi-xaw to provide /usr/bin/xdvi.bin

```



```

(xdvi.bin) in auto mode
update-alternatives: using /usr/bin/bibtex.original to provide /usr/bin/bibtex
(bibtex) in auto mode
Setting up lmodern (2.004.5-6.1) ...
Setting up texlive-base (2021.20220204-1) ...
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
/usr/bin/ucfr
mktexlsr: Updating /var/lib/texmf/ls-R-TEXLIVEDIST...
mktexlsr: Updating /var/lib/texmf/ls-R-TEXMFMAIN...
mktexlsr: Updating /var/lib/texmf/ls-R...
mktexlsr: Done.
tl-paper: setting paper size for dvips to a4:
/var/lib/texmf/dvips/config/config-paper.ps
tl-paper: setting paper size for dvipdfmx to a4:
/var/lib/texmf/dvipdfmx/dvipdfmx-paper.cfg
tl-paper: setting paper size for xdvi to a4: /var/lib/texmf/xdvi/XDvi-paper
tl-paper: setting paper size for pdftex to a4: /var/lib/texmf/tex/generic/tex-
ini-files/pdftexconfig.tex
Setting up tex-gyre (20180621-3.1) ...
Setting up texlive-plain-generic (2021.20220204-1) ...
Setting up texlive-latex-base (2021.20220204-1) ...
Setting up texlive-latex-recommended (2021.20220204-1) ...
Setting up texlive-pictures (2021.20220204-1) ...
Setting up texlive-fonts-recommended (2021.20220204-1) ...
Setting up tipa (2:1.3-21) ...
Setting up texlive (2021.20220204-1) ...
Setting up texlive-latex-extra (2021.20220204-1) ...
Setting up texlive-xetex (2021.20220204-1) ...
Setting up rake (13.0.6-2) ...
Setting up libruby3.0:amd64 (3.0.2-7ubuntu2.11) ...
Setting up ruby3.0 (3.0.2-7ubuntu2.11) ...
Setting up ruby (1:3.0~exp1) ...
Setting up ruby-rubygems (3.3.5-2ubuntu1.2) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for mailcap (3.70+nmu1ubuntu1) ...
Processing triggers for fontconfig (2.13.1-4.2ubuntu5) ...
Processing triggers for libc-bin (2.35-0ubuntu3.8) ...
/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc_proxy.so.2 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtcm.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libumf.so.1 is not a symbolic link

```

```

/sbin/ldconfig.real: /usr/local/lib/libur_loader.so.0 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_0.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libhwloc.so.15 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_level_zero_v2.so.0 is not a
symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbb.so.12 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind_2_5.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtcm_debug.so.1 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libtbbbind.so.3 is not a symbolic link

/sbin/ldconfig.real: /usr/local/lib/libur_adapter_opencl.so.0 is not a symbolic
link

/sbin/ldconfig.real: /usr/local/lib/libtbbmalloc.so.2 is not a symbolic link

Processing triggers for tex-common (6.17) ...
Running updmap-sys. This may take some time... done.
Running mktexlsr /var/lib/texmf ... done.
Building format(s) --all.
    This may take some time... done.
Collecting py pandoc
  Downloading py pandoc-1.16.2-py3-none-any.whl.metadata (16 kB)
  Downloading py pandoc-1.16.2-py3-none-any.whl (19 kB)
Installing collected packages: py pandoc
Successfully installed py pandoc-1.16.2
Collecting data science
  Downloading data science-0.18.1-py3-none-any.whl.metadata (910 bytes)
Requirement already satisfied: folium>=0.9.1 in /usr/local/lib/python3.12/dist-
packages (from data science) (0.20.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-
packages (from data science) (75.2.0)
Requirement already satisfied: matplotlib>=3.0.0 in
/usr/local/lib/python3.12/dist-packages (from data science) (3.10.0)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages
(from data science) (2.2.2)
Requirement already satisfied: scipy in /usr/local/lib/python3.12/dist-packages
(from data science) (1.16.3)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages
(from data science) (2.0.2)
Requirement already satisfied: ipython in /usr/local/lib/python3.12/dist-

```

packages (from datascience) (7.34.0)
 Requirement already satisfied: plotly in /usr/local/lib/python3.12/dist-packages
 (from datascience) (5.24.1)
 Requirement already satisfied: branca in /usr/local/lib/python3.12/dist-packages
 (from datascience) (0.8.2)
 Requirement already satisfied: jinja2>=2.9 in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (3.1.6)
 Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (2.32.4)
 Requirement already satisfied: xyzservices in /usr/local/lib/python3.12/dist-
 packages (from folium>=0.9.1->datascience) (2025.10.0)
 Requirement already satisfied: contourpy>=1.0.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (1.3.3)
 Requirement already satisfied: cycycler>=0.10 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.0.0->datascience) (0.12.1)
 Requirement already satisfied: fonttools>=4.22.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (4.60.1)
 Requirement already satisfied: kiwisolver>=1.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (1.4.9)
 Requirement already satisfied: packaging>=20.0 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (25.0)
 Requirement already satisfied: pillow>=8 in /usr/local/lib/python3.12/dist-
 packages (from matplotlib>=3.0.0->datascience) (11.3.0)
 Requirement already satisfied: pyparsing>=2.3.1 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (3.2.5)
 Requirement already satisfied: python-dateutil>=2.7 in
 /usr/local/lib/python3.12/dist-packages (from matplotlib>=3.0.0->datascience)
 (2.9.0.post0)
 Collecting jedi>=0.16 (from ipython->datascience)
 Downloading jedi-0.19.2-py2.py3-none-any.whl.metadata (22 kB)
 Requirement already satisfied: decorator in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (4.4.2)
 Requirement already satisfied: pickleshare in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.7.5)
 Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (5.7.1)
 Requirement already satisfied: prompt-toolkit!=3.0.0,!3.0.1,<3.1.0,>=2.0.0 in
 /usr/local/lib/python3.12/dist-packages (from ipython->datascience) (3.0.52)
 Requirement already satisfied: pygments in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (2.19.2)
 Requirement already satisfied: backcall in /usr/local/lib/python3.12/dist-
 packages (from ipython->datascience) (0.2.0)
 Requirement already satisfied: matplotlib-inline in

```

/usr/local/lib/python3.12/dist-packages (from ipython->datascience) (0.2.1)
Requirement already satisfied: pexpect>4.3 in /usr/local/lib/python3.12/dist-
packages (from ipython->datascience) (4.9.0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-
packages (from pandas->datascience) (2025.2)
Requirement already satisfied: tenacity>=6.2.0 in
/usr/local/lib/python3.12/dist-packages (from plotly->datascience) (9.1.2)
Requirement already satisfied: parso<0.9.0,>=0.8.4 in
/usr/local/lib/python3.12/dist-packages (from jedi>=0.16->ipython->datascience)
(0.8.5)
Requirement already satisfied: MarkupSafe>=2.0 in
/usr/local/lib/python3.12/dist-packages (from
jinja2>=2.9->folium>=0.9.1->datascience) (3.0.3)
Requirement already satisfied: ptyprocess>=0.5 in
/usr/local/lib/python3.12/dist-packages (from pexpect>4.3->ipython->datascience)
(0.7.0)
Requirement already satisfied: wcwidth in /usr/local/lib/python3.12/dist-
packages (from prompt-
toolkit!=3.0.0,!<3.0.1,<3.1.0,>=2.0.0->ipython->datascience) (0.2.14)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-
packages (from python-dateutil>=2.7->matplotlib>=3.0.0->datascience) (1.17.0)
Requirement already satisfied: charset_normalizer<4,>=2 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (3.4.4)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-
packages (from requests->folium>=0.9.1->datascience) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.12/dist-packages (from
requests->folium>=0.9.1->datascience) (2025.11.12)
Downloading datascience-0.18.1-py3-none-any.whl (725 kB)
725.1/725.1 kB
22.0 MB/s eta 0:00:00
Downloading jedi-0.19.2-py2.py3-none-any.whl (1.6 MB)
1.6/1.6 MB
60.7 MB/s eta 0:00:00
Installing collected packages: jedi, datascience
Successfully installed datascience-0.18.1 jedi-0.19.2

```

```

[2]: # Connect Google Drive to Colab so you can access your files
from google.colab import drive
drive.mount('/content/drive')

```

```
import os
os.chdir('/content/drive/MyDrive/Colab Notebooks/')
```

Mounted at /content/drive

```
[3]: # Run this cell, but please don't change it.

# These lines import the Numpy and Datascience modules.
import numpy as np
from datascience import *

# These lines do some fancy plotting magic.
import matplotlib
%matplotlib inline
import matplotlib.pyplot as plots
plots.style.use('fivethirtyeight')
import warnings
warnings.simplefilter('ignore', FutureWarning)
```

1 1. How Faithful is Old Faithful?

Old Faithful is a geyser in Yellowstone National Park that is famous for eruption on a fairly regular schedule.

```
[4]: # For the curious: this is how to display a YouTube video in a
# Jupyter notebook. The argument to YouTubeVideo is the part
# of the URL (called a "query parameter") that identifies the
# video. For example, the full URL for this video is:
# https://www.youtube.com/watch?v=wE8NDuzt8eg
from IPython.display import YouTubeVideo
YouTubeVideo("wE8NDuzt8eg")
```

[4]:



Some of Old Faithful's eruptions last longer than others. Whenever there is a long eruption, it is usually followed by an even longer wait before the next eruption. If you visit Yellowstone, you might want to predict when the next eruption will happen, so that you can see the rest of the park instead of waiting by the geyser.

Today, we will use a dataset on eruption durations and waiting times to see if we can make such predictions accurately with linear regression.

The dataset has one row for each observed eruption. It includes the following columns: - **duration**: Eruption duration, in minutes - **wait**: Time between this eruption and the next, also in minutes

Run the next cell to load the dataset.

```
[5]: faithful = Table.read_table("./DS/faithful.csv")
      faithful
```

```
[5]: duration | wait
      3.6      | 79
      1.8      | 54
      3.333    | 74
      2.283    | 62
      4.533    | 85
      2.883    | 55
```

```
4.7      | 88
3.6      | 85
1.95     | 51
4.35     | 85
... (262 rows omitted)
```

Question 1.0. The following statements are the unordered steps of linear regression.

1. Compute the parameters of the regression line: the slope and the intercept.
2. Evaluate the regression line by computing the line's RMSE and analyzing the residuals plot.
3. Use the regression line to generate predictions for each x value.
4. Determine if linear regression is a reasonable method by visualizing your data and computing the correlation coefficient.

Make an array called `least_squares_order` that contains the correct order of a linear regression analysis, where the first item of the array is the first step of an linear regression analysis and the last item of the array is the last step of an linear regression analysis.

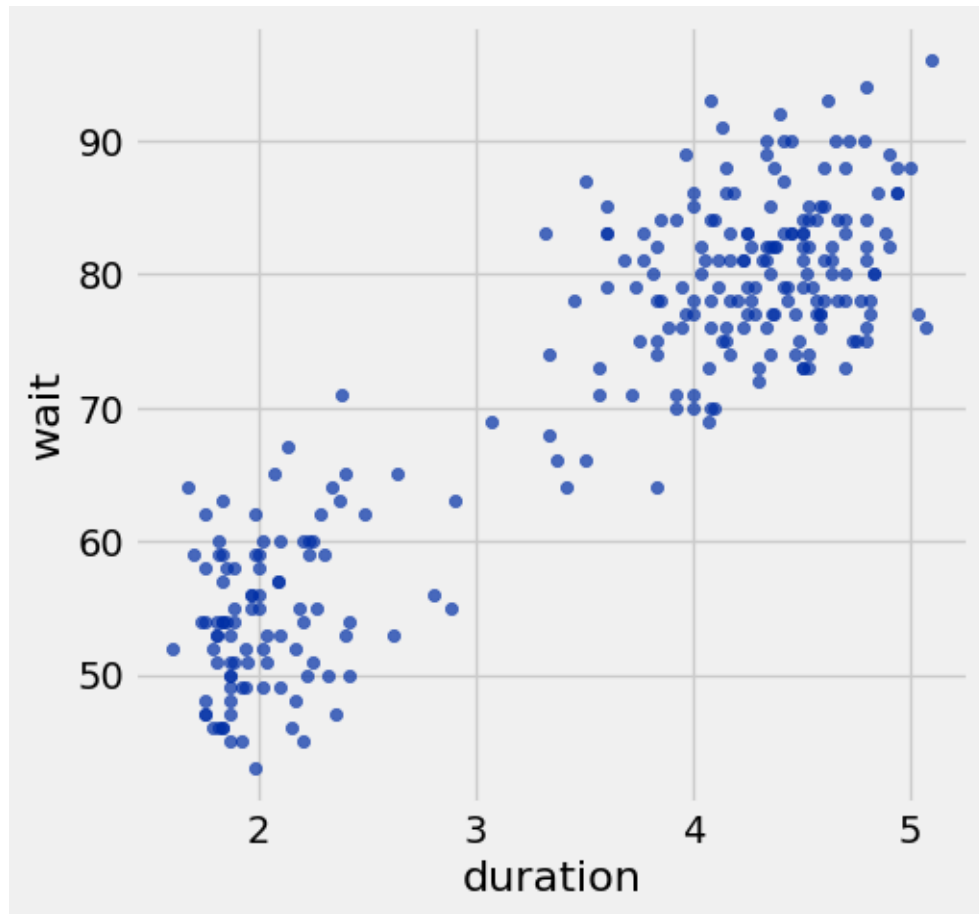
```
[7]: least_squares_order = make_array(4, 1, 3, 2)
least_squares_order
```

```
[7]: array([4, 1, 3, 2])
```

We would like to use linear regression to make predictions, but that won't work well if the data aren't roughly linearly related. To check that, we should look at the data.

Question 1.1. Make a scatter plot of the data. It's conventional to put the column we want to predict on the vertical axis and the other column on the horizontal axis.

```
[9]: faithful.scatter('duration', 'wait')
```



Question 1.2. Are eruption duration and waiting time roughly linearly related based on the scatter plot above? Is this relationship positive?

The scatter plot shows a similar relationship between eruption time and waiting time, and this relationship is positive. That is, the timing of an eruption tends to adjust the waiting time until the next eruption.

We're going to continue with the assumption that they are linearly related, so it's reasonable to use linear regression to analyze this data.

We'd next like to plot the data in standard units. If you don't remember the definition of standard units, textbook section [14.2](#) might help!

Question 1.3. Compute the mean and standard deviation of the eruption durations and waiting times. **Then** create a table called `faithful_standard` containing the eruption durations and waiting times in standard units. The columns should be named `duration (standard units)` and `wait (standard units)`.

```
[10]: duration_mean = faithful.column('duration').mean()
      duration_std = faithful.column('duration').std()
      wait_mean = faithful.column('wait').mean()
```



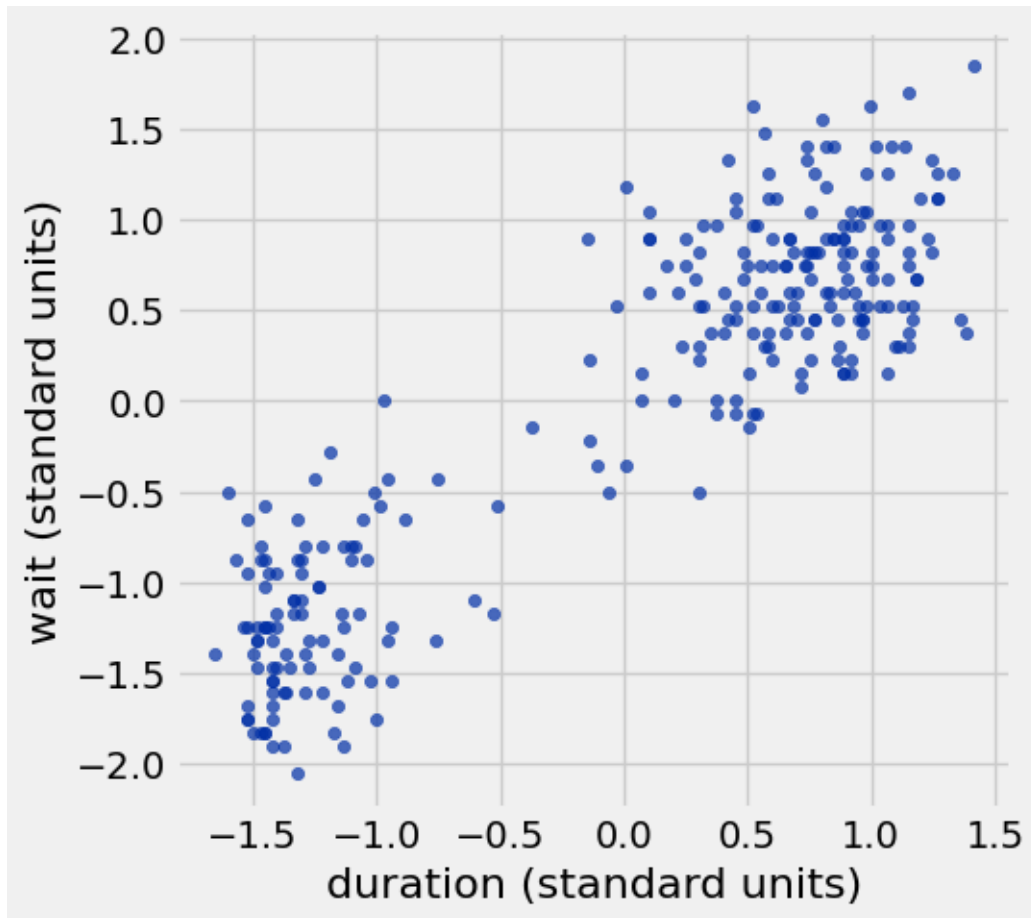
```
wait_std = faithful.column('wait').std()

faithful_standard = Table().with_columns(
    "duration (standard units)", (faithful.column('duration') - duration_mean) /
    ↪ duration_std,
    "wait (standard units)", (faithful.column('wait') - wait_mean) / wait_std)
faithful_standard
```

```
[10]: duration (standard units) | wait (standard units)
0.0984989                      | 0.597123
-1.48146                      | -1.24518
-0.135861                    | 0.228663
-1.0575                      | -0.655644
0.917443                     | 1.03928
-0.530851                    | -1.17149
1.06403                      | 1.26035
0.0984989                    | 1.03928
-1.3498                      | -1.46626
0.756814                     | 1.03928
... (262 rows omitted)
```

Question 1.4. Plot the data again, but this time in standard units.

```
[11]: faithful_standard.scatter('duration (standard units)', 'wait (standard units)')
```



You'll notice that this plot looks the same as the last one! However, the data and axes are scaled differently. So it's important to read the ticks on the axes.

Question 1.5. Among the following numbers, which would you guess is closest to the correlation between eruption duration and waiting time in this dataset?

1. -1
2. 0
3. 1

Assign `correlation` to the number corresponding to your guess (either 1, 2 or 3).

```
[12]: correlation = 3
      correlation
```

[12]: 3

Question 1.6. Compute the correlation coefficient: `r`.

Hint: Use `faithful_standard`. Section 15.1 explains how to do this.

```
[13]: r = (faithful_standard.column('duration (standard units)' * faithful_standard.  
      ↪column('wait (standard units)')).mean()  
      r
```

```
[13]: 0.90081116832181318
```

1.1 Submission

Make sure you have run all cells in your notebook in order before running the cell below, so that all images/graphs appear in the output. The cell below will generate a pdf file for you to submit. **Please save before exporting!**

```
[14]: # should change the directory and file name matching to yours  
      !jupyter nbconvert './DS/lab23(SSU).ipynb' --to pdf
```

```
[NbConvertApp] Converting notebook ./DS/lab23(SSU).ipynb to pdf  
[NbConvertApp] Support files will be in lab23(SSU)_files/  
[NbConvertApp] Making directory ./lab23(SSU)_files  
[NbConvertApp] Writing 65715 bytes to notebook.tex  
[NbConvertApp] Building PDF  
[NbConvertApp] Running xelatex 3 times: ['xelatex', 'notebook.tex', '-quiet']  
[NbConvertApp] Running bibtex 1 time: ['bibtex', 'notebook']  
[NbConvertApp] WARNING | bibtex had problems, most likely because there were no  
citations  
[NbConvertApp] PDF successfully created  
[NbConvertApp] Writing 134863 bytes to DS/lab23(SSU).pdf
```